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tl;dr:

ML Engineering Manager and AI Hacker, solving the hardest problems by delivering bleeding-edge research into production. I have worked for 20+ years as a Researcher, Engineer, and Manager, tackling problems at every level: real-time scale, frontier innovation, and shipping usable products. My teams and I have consistently solved complex problems, owned a ton of infrastructure, and celebrated the delivery of ambitious projects. These days I'm bringing closed-source-level performance to on-prem: fine-tuned open-weight models and evaluation stacks that run entirely inside the boundary, no data egress. Whatever my ikigai is, it braids Mathematics and Programming; lfg.

ML + AI LLMs, Generative AI, AI Agents, RAG, Fine-tuning, PyTorch, vLLM, Evals, Knowledge Graphs

Engineering + Platform Terraform, Kubernetes, MLOps, Airflow, Kafka, Celery, AWS, GCP, Golang, Python

Achievements

- Building Clearwing, an autonomous vulnerability scanner and source-code hunter - AI cyber security research
- Curated data collection and evaluation of unaligned models - covered in The Economist
- Fine-tuned Qwen3 for vulnerability scanning and tool usage
- Managed Backend and Data Engineering teams solving large-scale data problems
- Launched innovative, agent-based, llm-powered products
- Architected and deployed massive-scale, real-time data pipelines: 1.2M+ records/second
- Maintained infrastructure comprising 1,000+ instances
- Freelanced as a Founding AI Engineer, building ML applications for seed-stage companies
- Fine-tuned Mixtral to build red-team models for synthetic data generation

Education Bachelor of Science, Mathematics - Boston University

Work Experience

Lazarus AI — **ML Engineering Manager** — *October 2025 - Present*

Leading ML engineering and applied research across a broad portfolio: AI-driven exploit analysis and penetration testing, document extraction, on-premise contract-driven projects, and confidential computing. Consulting to core engineering on a per-project basis, bringing bleeding-edge research into production systems.

- Drove evaluation-driven engineering culture, improving document extraction correctness from 20% to 85%
- Demonstrated end-to-end AI-driven vulnerability scanning to RCE using self-hosted unaligned open-weight models - Clearwing
- Directed four concurrent project lines as ML Engineering Manager: on-prem privacy-focused public-sector delivery, 1,000+-page insurance underwriting documents, evaluation of novel unaligned fine-tunes, and the Clearwing adversarial security harness
- Managed ML team with rotating contributors, providing technical direction and per-project leadership to core engineering
- Built evaluation harnesses across document extraction, agentic task performance, and CVE exploit validation: LLM-as-judge, golden datasets, offline + CI benchmarking - Langfuse, Python
- Overhauled LLM observability to run fully on-prem under HIPAA and SOC 2 constraints - zero third-party data egress: trace propagation, cost tracking, and evals - Langfuse, k3s, HIPAA, SOC 2
- Shipped features for Clearwing, an autonomous vulnerability research agent: sandboxed analysis, exploit validation, budget-bounded orchestration - Python, Docker
- Fine-tuned and deployed open-weight models on self-hosted inference infrastructure - vLLM, LoRA, k3s
- Owned data collection and evaluation for ReAligned, a Qwen3.5 finetune eliminating propaganda, lying, and gaslighting: 0.8B - 35B public, plus privately-released 122B and 397B variants; covered in The Economist - Qwen3.5
- Architected multi-model routing layer with confidence scoring, OCR grounding, and post-inference validation - LangGraph
- Owned production ML platform: k3s cluster, ArgoCD, LiteLLM, Dagster pipelines - Terraform, Kubernetes

Warmer — **Founding AI Engineer** — *November 2024 - October 2025*

Building the future of Financial Advisory. In this role, I launched two AI products from zero to production: a financial advisor assistant and intelligent client-advisor matching system, optimizing Cost-per-Lead.

- Zero-to-One: founded and scaled AI product from zero to first paying customer - LangGraph, OpenAI, Python
- Built conversational, lead-gen agent, the foundation of the new business model - LangGraph, LLM

- Achieved <1% hallucinations on golden-set evals through grounding and post-inference validation - LangGraph, LangSmith
- Architected multi-agent system managing complete client lifecycle, processing 1000+ meetings per day - LangGraph
- Introduced experiment-driven development culture with A/B testing and feature flags - PostHog
- Built sub-12 second transcript analysis pipeline for real-time processing - Celery, Amazon SQS
- Designed complete agent architecture including RAG, memory, and tool usage systems - pgvector, RAG, LangGraph
- Implemented hybrid-search RAG system across 1,000,000+ client communications - GraphRAG/LightRAG, pgvector
- Modified open-source dynamic Knowledge Graph, LightRAG, to support pgvector/Postgres backend - pgvector, Python

AI Consulting — **AI Consultant and Contractor** — *September 2023 - November 2024*

Contracted with several early-stage startups, building out AI products.

WorkMate Labs: Workmate is your AI teammate

- Designed and built a Digital Assistant, automating task extraction and email drafting - LLM, TypeScript
- Designed and deployed LLM Prompts for contextual, personal email drafting - ChatGPT, Gemini, LLM
- Designed and deployed custom evaluation framework for measuring LLM performance - LangSmith, TypeScript
- Designed retrieval system for context-aware search of similar sent emails - Retrieval, TypeScript

Stealth Startup: FoodTech, translating conversations into API calls

- Fine-tuned Dolphin Mixtral to improve conversational style - Mixtral, HuggingFace, Python
- Designed and built question-answering workflows inspired by Graph-of-thoughts - OpenAI API, Python
- Designed and built SMS-based chat interface to LLMs - Amazon Bedrock, LangChain, Twilio, Python
- Automated MLOps for LLM-based projects - finetuning, testing, deployment, and performance evaluation
- Designed and deployed IaC for all services - Terraform

Stealth Startup: Generative AI Customers as a service

- Diagnosed and solved a mission-critical problem for scheduling flow of traffic to clients - Math!
- Designed and built API to develop, simulate, and actually orchestrate system traffic flows - Python
- Standardized MLOps approach for use with Stable Diffusion and LLMs - Terraform
- Designed and built tooling to monitor, compare, and research LLM performance - Python, LLM, LLaMa

Owl.co — **Software Engineer** — *January 2023 - September 2023*

Returning to the startup world, I designed and built products, catering to the Insurance Industry, with a focus on integrating Machine Learning with Human tasks. Here my focus was on ML products instead of Data scale.

- Led an interdisciplinary team building ML products, automating tasks performed by human investigators: scraping the web, classifying documents, designing ETLs - Clojure, PyTorch, Presto, Spark
- Architected systems to integrate ML inference with human-driven tasks - Clojure, Amazon SageMaker
- Automated hundreds of daily insurance claims investigations with the use of ML, reducing manual investigations by 70%
- Designed and built ETL workflows for analytic databases - Airflow, Spark
- Built dashboards tracking performance of ML models against their human counterparts - Presto, Airflow
- Deployed and maintained infrastructure - AWS CloudFormation, Amazon SageMaker
- Mentored mid-level Engineers on Data Engineering

Sabbatical — **Career Break** — *October 2021 - January 2023*

- Traded the post-acquisition golden handcuffs for plane tickets
- Came back deliberately, trading scale for deep focus in AI

Oracle Data Cloud, MOAT — **Software Engineering Manager** — *December 2019 - October 2021*

I led a highly technical team to create a vast, event-level data store, used as the source-of-truth for the suite of MOAT products. The real-time system processes 1.2M+ records/second, and requires zero downtime. Consequently, I grew a team with high technical aptitude, and emphasized ownership as a core principle in Software Development.

- Managed and grew team of 7 Data Engineers, ranging from College Recruit to Senior Engineer
- Built multiyear Software Roadmap with Engineering Managers and Product Owners
- Mentored and promoted every Software Engineer on my team
- Collaborated with ML Engineers and Data Scientists to release and update models in production code
- Collaborated with outside Engineering and Data Science stakeholders to design a flexible data pipeline
- Led project to migrate legacy systems from EC2 to Kubernetes (EKS) - Kubernetes
- Migrated legacy core business logic to modern systems - Kafka, Airflow
- Managed a team owning 1,000+ instances - AWS
- Managed a budget of $340,000+ per month
- Authored technical proposals for Data Privacy, System Architecture, and Wire Protocols
- Co-wrote and presented software application proposals, detailing and defending technology decisions
- Reviewed and approved technical design proposals and outage postmortems

Oracle Data Cloud, MOAT — **Tech Lead, Data Engineering** — *February 2017 - December 2019*

I stabilized and scaled a massive computing cluster, halved instance count, and saved over $2M annually. Comprising a massive 30k-line codebase, the real-time system contained all business logic to power the MOAT dashboard, and required biweekly deployments. Here, I emphasized stability and correctness, deploying frequent changes across 1,000+ instances.

- Managed weekly software releases for core business logic, contributed to by 4 distinct teams
- Built multiyear roadmap for the data pipeline, and the systems that power it
- Onboarded all new hires to MOAT's data pipeline
- Designed and built stream-processing applications processing 1.2M+ events/second - Golang, Python, Kafka
- Designed and built system-wide wire protocol - Protobuf
- Built custom software that reduced instance count by 50%, saving over $2M dollars - Golang
- Built and maintained software end-to-end over 1,000+ AWS instances (c5.xl, r5.8xl)
- Designed "cold storage" data schema - Parquet
- Maintained historical databases, importing 800,000,000+ rows per day - Highly modified Postgres
- Acquired by Oracle Data Cloud

Chartbeat — **Tech Lead, Data Engineering** — *December 2014 - December 2016*

I led an interdisciplinary team as a product-minded Data Engineer, building both the core data pipeline and an initial version of the Chartbeat Historical product. This position introduced me to large-scale distributed systems, leadership, and implementing product-facing changes.

- Led 7-person interdisciplinary Scrum Team
- Designed and built core data pipeline, processing 300,000+ events per second - Kafka and Clojure
- Designed and maintained session-level data warehouse - Amazon Redshift
- Designed and maintained sub-second query databases, importing 1,000,000+ rows per hour - Postgres
- Designed wire protocol - Protobuf
- Built and maintained real-time data-scrubbing libraries - Clojure, Java
- Deployed and configured production machines - Puppet, Fabric

Harvard University, IQSS — **Statistical Programmer, Software Maintainer** — *May 2010 - July 2013*

Statistical software and data privacy research at Harvard's Institute for Quantitative Social Science. Where I learned to turn research ideas into production, open-source code.

- Developed and maintained Zelig, an open-source statistical package unifying model interfaces - R
- Contributed statistical analysis software to Dataverse, open-source infrastructure for research data sharing, preservation, and citation - Java, R
- Built the Zelig extension to the Dataverse network - Java
- Taught statistical programming workshops - beginner through developing statistical packages